# Electrical Safety and LOTO

## Voice Over Script Document

# Course Title

**Electrical Safety and Lockout-Tagout (LOTO)**

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# Course Overview

Electricity powers critical operations across pharmaceutical manufacturing facilities, from granulation and compression to coating and packaging. While essential for productivity and quality, electrical energy can also create serious hazards if not properly controlled.

This course provides employees with essential knowledge on electrical hazards, hazardous area classification, electrical protective systems, safe work practices, and Lockout-Tagout (LOTO) procedures used in pharmaceutical operations.

Learners will understand how to identify risks, protect themselves from electrical hazards, and safely isolate hazardous energy sources before performing maintenance or servicing activities.

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# Course Outline

| Module | Topic |
| --- | --- |
| Module 1 | Introduction to Electrical Safety and LOTO |

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| Module 2 | Electrical Hazards and Prevention |
| Module 3 | Hazardous Area Classification |
| Module 4 | Electrical Safety Controls and Safe Work Practices |
| Module 5 | Lockout-Tagout (LOTO) |
| Module 6 | Assessment and Course Completion |

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# Module 1: Introduction to Electrical Safety and LOTO

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## 1.1 Welcome

### Voice Over Script

Welcome to the course on “Electrical Safety and LOTO.”

Everyday electricity powers process operations in API and Formulation units—from granulation to compression and coating.

When managed correctly, electricity supports safe, reliable, and efficient operations.

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### On-Screen Text

- Electrical Safety and LOTO
- Safe • Reliable • Efficient Operations

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## 1.2 Quick Facts and Impacts

### Voice Over Script

Did you know that in India, nearly 12 people die each day from electrocution?

Almost 42% of industrial fires originate from electrical faults.

And 8% of all factory deaths are caused by electricity.

At Granules India, we handle sensitive pharmaceutical materials and precision machinery. A single spark can harm people and contaminate products.

This is why electrical safety isn't optional—it's a culture of care.

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### On-Screen Text

- 12 deaths/day from electrocution
- 42% industrial fires from electrical faults
- Electrical safety is a culture of care

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## 1.3 Course Objective

### Voice Over Script

By the end of this course, you'll be able to:

- Identify electrical hazards and their effects on people and processes.
- Recognize hazardous area classifications in pharma operations.
- Apply safe work practices and protective controls.
- Follow the correct Lockout-Tagout procedure for maintenance and cleaning tasks.

Click on the cards to access each module.

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### On-Screen Text

By the end of this course, you will be able to:

- Identify electrical hazards
- Recognize hazardous areas
- Apply safe work practices
- Follow LOTO procedures

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# Module 2: Electrical Hazards and Prevention

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## 2.1 Our Power Infrastructure

### Voice Over Script

To work safely, we must first understand the scale of energy powering our facility.

We manage a high-voltage distribution system that begins at 33kV, stepping down to 11kV before reaching our Power Control Centers, or PCCs.

From there, electricity is fed to Motor Control Centers, or MCCs, that drive our production lines.

Because this system carries the power required to run an entire manufacturing site, it demands our highest level of professional respect and strict adherence to safety protocols.

This infrastructure is designed to be highly controlled and safe.

However, if that energy escapes its intended path—due to equipment failure or human error—it manifests in three ways: Shock, Burn, or Arc Flash.

Let’s look at these electrical risks.

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### On-Screen Text

- 33kV → 11kV → PCC → MCC
- Shock
- Burn
- Arc Flash

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# 2.2 What is Electric Shock?

## Voice Over Script

An electric shock occurs when electrical current passes through the human body and completes a circuit.

In simple terms, your body becomes a pathway for electricity to flow.

The result can range from a mild tingling sensation to severe burns, heart failure, or even death.

The severity of shock depends on three critical factors:

- Path
- Amount of current
- Duration

Click on each circle to know more.

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## Interactive Elements

### Path

Current passing through vital organs such as the heart, lungs, or brain can be deadly.

A hand-to-hand current path crosses the chest and heart, making it one of the most dangerous paths.

### Amount of Current

- More than 3 mA — Painful shock
- More than 10 mA — “No Let Go” danger
- More than 30 mA — Lung paralysis

- More than 50 mA — Ventricular fibrillation
- 100 mA to 4 A — Fatal fibrillation
- Over 4 A — Severe burns and heart paralysis

**Duration**

The longer the contact with electricity, the greater the damage.

Muscles can lock onto the energized source, increasing exposure time and risk.

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### On-Screen Text

Electric Shock Severity Depends On:

- Path
- Current
- Duration

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## 2.3 Burns, Arc Flash & Arc Blast

### Voice Over Script

Working on or near electrical equipment and panels involves serious hazards.

Our PCCs, MCCs, and other electrical systems carry high energy levels.

If this energy is suddenly released due to a fault, equipment failure, or human error, it can cause severe injuries in the form of burns, arc flash, or arc blast.

Let us understand these hazards.

Electrical Burns:
Intense heat causes deep tissue damage, often affecting the hands during maintenance work.

Arc Flash:
An arc flash occurs when high-amperage current jumps through the air between conductors or to ground.

Temperatures can reach up to 30,000 degrees Celsius—four times hotter than the surface of the sun.

Arc Blast:
The rapid expansion of air creates a pressure wave powerful enough to rupture eardrums, throw workers across rooms, and turn metal fragments into deadly projectiles.

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### On-Screen Text

- Electrical Burns
- Arc Flash
- Arc Blast
- Up to 30,000°C

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## 2.4 Effects of Electrical Hazards

### Voice Over Script

Electrical hazards can cause serious harm.

Their effects may be immediate or long-term and can impact people, operations, and the organization.

Electrical incidents can result in:

- Serious injury or loss of life
- Burns, nerve injury, or loss of vision
- Falls and secondary injuries
- Mental stress and trauma

Operational impacts include:

- Production shutdowns
- Damage to PCCs and MCCs
- Product contamination risks
- Downtime and delays

Organizational impacts include:

- Legal consequences
- Financial losses
- Increased inspections and restrictions

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### On-Screen Text

Electrical Incidents Impact:

- People
- Operations
- Organization

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## 2.5 Secondary Injuries

### Voice Over Script

Not every electrical accident ends with a visible burn or shock.

Sometimes the indirect effects are just as serious.

A sudden shock can cause a worker to lose balance, fall from height, or drop tools and equipment.

Secondary injuries can also result from:

- Striking nearby machinery
- Product spills or contamination
- Uncontrolled movements

Always use non-conductive ladders, maintain safe clearances, and ensure dry footing in damp or elevated locations.

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### On-Screen Text

Prevent Secondary Injuries:

- Use non-conductive ladders

- Maintain clearances
- Ensure dry footing

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## 2.6 Precautionary Measures

### Voice Over Script

While engineering controls are in place, your last line of defense is the equipment you wear.

To stay safe, follow strict PPE and location-specific protocols.

For Head:
Use insulated non-conductive hard hats.

For Hands:
Use insulated rubber gloves.

For Feet:
Wear dielectric footwear.

For Body:
Wear fire-resistant clothing to protect against arc flash.

In transformer yards:

- Access is restricted to authorized personnel.
- Maintain safe approach boundaries.

In PCC and MCC rooms:

- Verify Lockout-Tagout before work.
- Ensure insulating rubber mats are dry.
- Use 1,000-volt rated insulated tools.
- Stand to the side when operating breakers.

By following these precautions, you help ensure a safe workplace.

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### On-Screen Text

Required PPE:

- Hard Hat
- Rubber Gloves
- Dielectric Footwear
- Arc-Rated Clothing

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# 2.7 Preventing Electrical Hazards

### Voice Over Script

Electrical hazards are dangerous—but preventable.

Always:

- De-energize equipment before work.
- Inspect cables, plugs, and tools before use.
- Use correctly rated electrical equipment.
- Keep the workplace dry.
- Avoid electrical overloads.
- Use PPE and insulated tools.
- Follow warning signs and barriers.

Practicing these basics every day keeps people, processes, and products safe.

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### On-Screen Text

Prevent Electrical Hazards:

- De-energize
- Inspect
- Avoid overloads
- Use PPE
- Follow warning signs

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# Module 3: Hazardous Area Classification

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## 3.1 Understanding Hazardous Areas

### Voice Over Script

In our pharmaceutical facility, we don't just look at where chemicals are—we look at how they behave.

We use two primary standards to define hazardous areas:

- Petroleum Rules 1976
- IS 5572

A location becomes hazardous when:

- Flammable vapors are present in ignitable concentrations.
- Heated liquids produce ignitable vapors.

IS 5572 helps ensure that electrical systems are designed to prevent sparks from igniting explosive atmospheres.

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### On-Screen Text

Hazardous Area Standards:

- Petroleum Rules 1976
- IS 5572

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## 3.2 Why Hazardous Area Classification Matters

### Voice Over Script

Electrical sparks, arcs, or heat can ignite solvent vapors.

Hazardous Area Classification helps prevent ignition by categorizing areas based on the likelihood and duration of explosive atmospheres.

This classification determines:

- Electrical system design
- Enclosure type
- Cable selection
- Maintenance practices

Common classified areas include:

- Solvent recovery tanks
- Granulation rooms
- Coating areas
- Exhaust ducts and trenches

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### On-Screen Text

HAC Helps Prevent:

- Ignition
- Explosions
- Fires

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## 3.3 Zone Classification as per IS 5572

### Voice Over Script

IS 5572 classifies hazardous areas into zones based on how often explosive gas, vapor, or combustible dust may be present.

Click on each zone to learn more.

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### Interactive Elements

**Zone 0**

Explosive atmosphere continuously present.

**Zone 1**

Explosive atmosphere likely during normal operation.

**Zone 2**

Explosive atmosphere unlikely and short duration only.

**Zone 20**

Combustible dust continuously present.

**Zone 21**

Combustible dust likely during normal operation.

**Zone 22**

Combustible dust unlikely and temporary.

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### On-Screen Text

Zone Classifications:

- Zone 0 / 1 / 2
- Zone 20 / 21 / 22

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## 3.4 How We Identify Zones in the Plant

### Voice Over Script

During site classification, engineers assess:

- Type of solvent or vapor
- Sources of release
- Ventilation efficiency
- Topography and vapor accumulation

These factors determine whether standard or explosion-proof electrical equipment is required.

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### On-Screen Text

Zone Identification Factors:

- Solvents
- Release Sources
- Ventilation
- Topography

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# Module 4: Electrical Safety Controls and Safe Work Practices

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## 4.1 Electrical Protective Devices

### Voice Over Script

Electrical protective devices act as the first line of defense against overloads, short circuits, and leakage currents.

These include:

- Fuses
- Circuit breakers
- Ground Fault Circuit Interrupters, or GFCIs

They protect both equipment and personnel by interrupting unsafe electrical conditions.

Never repeatedly reset a tripping breaker. A tripped breaker signals an underlying problem that must be investigated.

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### On-Screen Text

Protective Devices:

- Fuses
- Circuit Breakers
- GFCIs

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## 4.2 Grounding and Bonding

### Voice Over Script

Grounding and bonding help safely control unwanted electrical energy and static charges.

Grounding provides a low-resistance path to earth for fault current.

Bonding keeps metallic parts at the same electrical potential to prevent dangerous voltage differences.

In pharmaceutical facilities, grounding and bonding also help prevent electrostatic discharge during powder handling and solvent operations.

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### On-Screen Text

Grounding:
Safe path to earth

Bonding:
Equal electrical potential

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## 4.3 Guarding

### Voice Over Script

All live electrical parts operating at 50 volts or more must be guarded or enclosed to prevent accidental contact.

Electrical panels and MCCs must only be accessible to qualified personnel.

Circuit breakers must clearly indicate ON and OFF positions, and high-voltage warning signs must always be displayed.

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### On-Screen Text

Electrical Safety:

- Guard live parts
- Restrict access
- Display warning signs

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## 4.4 Illumination and Safe Access

### Voice Over Script

Adequate illumination must be provided around electrical panels and service equipment.

To ensure safe access:

- Maintain one-meter clearance in front of panels.
- Provide sufficient headroom.
- Keep panel areas free from obstructions.

Proper workspace discipline significantly reduces accidents.

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### On-Screen Text

Safe Access Requirements:

- 1-meter clearance
- Adequate lighting
- Obstruction-free panels

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## 4.5 Safe Work Practices

### Voice Over Script

Safe behavior is as important as engineering controls.

Always:

- Use insulated tools.
- Remove conductive jewelry.
- Treat untagged equipment as energized.
- Use warning signs and barricades.
- Assign safety observers where necessary.
- Coordinate with operations before electrical isolation.

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### On-Screen Text

Safe Work Practices:

- Use insulated tools
- Remove jewelry
- Use barricades
- Coordinate isolation

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# Module 5: Lockout-Tagout (LOTO)

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## 5.1 What is Lockout-Tagout?

### Voice Over Script

Lockout-Tagout, or LOTO, is a hazard control technique used to keep equipment in a safe, de-energized condition during servicing or maintenance.

Lockout physically secures an energy-isolating device in the OFF or SAFE position using a lock.

This prevents accidental or unauthorized operation.

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### On-Screen Text

LOTO:

- Isolate Energy
- Lock Equipment
- Prevent Startup

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## 5.2 Why Do We Need LOTO?

### Voice Over Script

During maintenance activities, workers face risks from:

- Unexpected energization
- Accidental startup
- Release of stored energy

LOTO prevents these failures by:

- Shutting down equipment
- Isolating energy sources
- Preventing unexpected release
- Verifying zero energy before work

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### On-Screen Text

LOTO Prevents:

- Unexpected startup
- Energy release
- Injuries and fatalities

## 5.3 The Fatal Five – Top Causes of LOTO Injuries

### Voice Over Script

The top causes of LOTO injuries include:

- Not stopping equipment
- Not disconnecting energy
- Not releasing stored energy
- Accidental restarting
- Not clearing the work area

All are preventable with proper LOTO procedures.

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### On-Screen Text

The Fatal Five:

- No shutdown
- No isolation
- Stored energy
- Restarting
- Unsafe area

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## 5.4 Types of Hazardous Energies

### Voice Over Script

Hazardous energy exists in many forms:

- Mechanical
- Electrical
- Thermal
- Chemical
- Hydraulic
- Pneumatic

Each type requires specific isolation methods and safety controls.

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### On-Screen Text

Hazardous Energy Types:

- Mechanical
- Electrical
- Thermal
- Chemical
- Hydraulic
- Pneumatic

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## 5.5 Energy-Isolating Devices

### Voice Over Script

Energy-isolating devices physically prevent energy from reaching equipment.

Common devices include:

- Disconnect switches
- Circuit breakers
- Line valves
- Blind flanges

Remember:
Stop buttons and interlocks do not isolate energy.

Always lock out the primary power source.

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### On-Screen Text

Energy-Isolating Devices:

- Disconnect Switches
- Circuit Breakers

- Valves
- Blind Flanges

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## 5.6 Isolation Techniques for Hazardous Energy

### Voice Over Script

Different hazardous energy types require different isolation techniques.

Electrical:
Use disconnect switches and breakers.

Mechanical:
Block or secure moving parts.

Hydraulic and Pneumatic:
Close valves and release pressure.

Thermal:
Allow systems to cool safely.

Chemical:
Close feed valves and neutralize hazards.

Gravity:
Use blocks or supports to prevent movement.

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### On-Screen Text

Isolation Techniques:

- Disconnect
- Block
- Vent
- Cool
- Neutralize
- Support

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# 5.7 Basic Steps of the LOTO Procedure

### Voice Over Script

The LOTO procedure follows a standardized 9-step sequence:

1. Identify all energy sources.
2. Notify affected employees.
3. Shut down equipment.
4. Isolate all energy sources.
5. Apply locks and tags.
6. Release stored energy.
7. Verify isolation.
8. Perform servicing tasks.
9. Remove locks safely and restore operations.

Verification is the most critical step.

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### On-Screen Text

9 Steps of LOTO

1. Identify
2. Notify
3. Shutdown
4. Isolate
5. Lock & Tag
6. Release Energy
7. Verify
8. Service
9. Restore

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# 5.8 Typical LOTO Points

### Voice Over Script

LOTO points are locations where energy is isolated to prevent equipment startup during maintenance.

Common LOTO points include:

- Emergency push buttons
- Electrical isolation switches
- Pneumatic air lines

Always verify that all identified energy sources are properly isolated before beginning work.

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### On-Screen Text

Common LOTO Points:

- Emergency Push Button
- Main Isolation Switch
- Pneumatic Line

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## 5.9 Key Takeaway

### Voice Over Script

LOTO keeps equipment in a safe, zero-energy state.

All hazardous energy sources must be identified, isolated, and verified before servicing.

Always follow the 9-step LOTO procedure and verify isolation before starting work.

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### On-Screen Text

Key Takeaways:

- Identify hazards
- Isolate energy
- Verify zero energy

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# Module 6: Assessment and Course Completion

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## Assessment

### Voice Over Script

To complete the course, you must take the final assessment.

The passing score is 80%.

The assessment contains 10 questions, and you have unlimited attempts.

During the assessment, you will not be able to access other course sections.

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### On-Screen Text

Assessment:

- 10 Questions
- Passing Score: 80%
- Unlimited Attempts

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## Course Completion

### Voice Over Script

You have successfully completed this course on Electrical Safety and LOTO.

Click the Close button to exit the course.

Thank you, and stay safe.

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### On-Screen Text

Congratulations!
You have completed:
Electrical Safety and LOTO